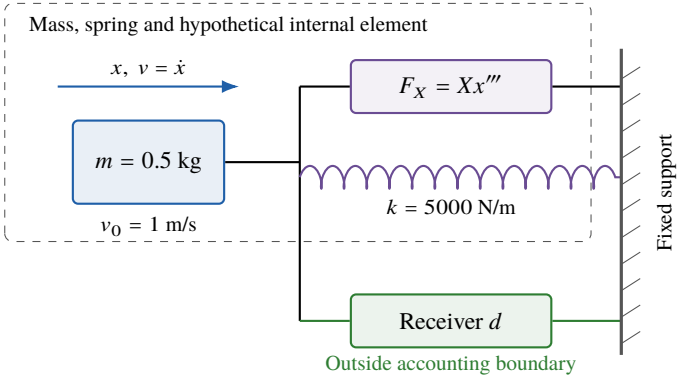

Forces on the mass

$-kx, \quad -dv, \quad -F_X$

Receiver port

$F_R = dv, \quad P_R = F_R v$

$W_R = \int F_R \, dx$

Stop at first $v = 0$